



$$\log_2 2 = 1$$

$$2^1 = 2$$

$$\log_2 4 = 2$$

$$2^2 = 4$$

$$\log_2 8 = 3$$

$$2^3 = 8$$

$$\log_2 16 = 4$$

$$2^4 = 16$$

$$\log_2 32 = 5$$

$$2^5 = 32$$

- 4.) Algorithm A uses $10n \log n$ operations, while algorithm B uses n^2 operations. Determine the value of n_0 such that A is better than B for $n \geq n_0$. Show your work.

$$10(10) \log_2 10 \approx 301.1928 \quad X$$

$$10^2 = 100$$

$$10(20) \log_2 20 \approx 864.385$$

$$20^2 = 400$$

$$10(30) \log_2 30 \approx 1472.6673$$

$$30^2 = 900$$

$$10(40) \log_2 40 \approx 2188.772$$

$$40^2 = 1600$$

$$10(50) \log_2 50 \approx 2821.9281$$

$$50^2 = 2500$$

$$10(59) \log_2 59 \approx 3463.51476$$

$$59^2 = 3481$$

$$\text{Answer} = n_0 \geq 59$$

- 5.) Determine which of the following formulas is $O(n)$:

a.) $16n^3$

b.) $n^2/2$

c.) $n^2 + n + 2$

d.) $10n + 25$

The answer is d since $O(n)$ refers to a linear relationship and $10n + 25$ represents that.